

Project summary

An agent runtime for prediction markets

Token2049 Origins hackathon. Status on 5 October 2026, from the *main* branch on GitHub. This version replaces the summary of 3 October 2026.

The project in one minute. Think of a robot trader with a strict rule book. A person writes the rules. The robot looks for good bets on future events. It buys the information that it needs, and it places bets. The rule book stops the robot when it tries to break a rule. A screen shows the person each thing that the robot does. We build the software for that robot. At this time, the robot uses only practice money.

What the project is about

A **prediction market** is a website where people buy and sell bets on future events. An example is "Will the US central bank change interest rates in October?" The price of a bet shows how likely the crowd thinks the event is.

We build software that runs a trading **agent** for prediction markets. An agent is a program that acts on its own for a person.

The person who controls the agent is the **operator**. The operator does not choose each trade. The operator writes the rules, starts the agent, and watches it. We call the rules the **policy**. The agent finds trades, pays for the data that it needs, and places orders. The policy stops the agent when it tries to do something that the rules do not permit.

Why this is useful. An agent can watch hundreds of markets at all hours. But an agent without limits can lose money fast. Our software gives the agent hard limits, a record of each decision, and a way to pay for data by itself.

River Markets helps human traders use many prediction market sites from one place. We give the same help to agents. We do not earn money from bets that win. We supply the software that runs the agent. Users pay each time they use it and for each order.

Key terms

You do not need to know about crypto or software to read this summary. These tables explain the words.

Market terms

Term	Meaning
Share	One unit of a bet. Each market has a "yes" side and a "no" side. A share pays \$1 if its side is correct and \$0 if it is not. A price of \$0.83 means that the market gives the event an 83% chance.
Venue	A website that runs prediction markets. We use Polymarket and Kalshi.
Order	An instruction to buy or sell shares. A paper order is a practice order. No real money goes to a venue. A live order uses real money.
Position	The shares that the agent holds in one market.
Profit and loss (P&L)	How much money the agent made or lost.
Resolution	The event happens, and the market pays the winning shares.

Crypto terms

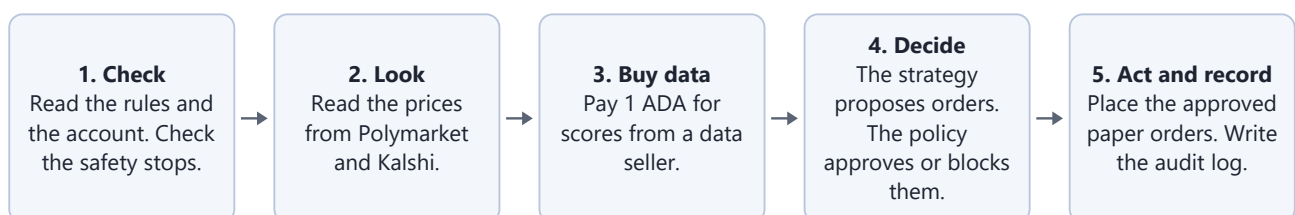
Term	Meaning
Blockchain	A public record of payments. Many computers keep copies of it, so no single company controls it.
Cardano and ADA	Cardano is the blockchain that our agent uses to pay for data. ADA is the money on Cardano.
Wallet	The account that holds the agent's ADA.
Escrow	A safe middle place for a payment. The money stays there until the seller delivers. If the seller does not deliver, the buyer can get a refund.
Masumi	A Cardano service that gives the agent an identity and holds each payment in escrow.
Test network (preprod)	A copy of Cardano for testing. Its ADA is free and has no real value.
x402	A method that lets a program pay for a web service. The service replies "payment required" and gives a price. The program pays, then asks again with proof of payment.
Chainlink CRE	A service that runs our main program on many independent computers. The computers must agree on each result. This makes the results hard to fake.

Software terms

Term	Meaning
Service	A program that runs all the time and answers requests from other programs.
API	The set of requests that a service accepts. Programs use it to talk to each other.
Database	A file that keeps data safe when a program stops and starts again.
Audit log	A record of each step and each decision that the agent makes.
UI	User interface. The screen that a person uses.
Automatic test	A small program that checks that another program does the correct thing.
GitHub	A website that keeps the project code and the history of each change.
Pull request	A request to add a change to the official version of the code. The team can see the change and its test results before they accept it.

How the agent works: one cycle

The agent works in cycles. A timer starts each cycle. This picture shows the five stages:



If a safety stop is on, the cycle ends after stage 1. The agent buys no data and places no orders.

In more detail, the agent does these steps in each cycle:

- 1 It reads the policy and the account balance.
- 2 It checks the safety stops. If a stop is on, the cycle ends there.
- 3 It reads the open markets from each venue that the policy permits. If one venue does not reply, the agent continues with the other venue.
- 4 It checks whether any market can pass the policy. If no market can pass, the agent does not buy data. This saves money.
- 5 It buys **scores** from a data seller. A score is the seller's estimate of the true chance of each event. The agent pays on Cardano with x402.
- 6 The **strategy pack** compares each price with its score. The strategy pack is the part of the agent that selects trades. If a share costs at least 4 cents less than the score, the strategy pack proposes an order.
- 7 The policy checks each proposed order. It approves or blocks the order and records the reason. It approves at most 5 orders in one cycle.
- 8 The agent sends the approved orders. Each order has its own key. If the agent sends the same order again, it does not buy twice.
- 9 The agent writes each step to the audit log.

Example from a test run. The score for "Will there be no change in Fed interest rates after the October 2026 meeting?" was higher than the price. The agent bought 24 "yes" shares at \$0.83 each, as a paper order.

The rules (policy)

The policy has these rules. The default value of each rule is in parentheses.

- **Maximum order:** the most money for one order (\$25).
- **Daily loss limit:** the most money the agent can lose in one day (\$100).
- **Categories:** the market topics the agent can or cannot trade ("politics" is blocked).
- **Venues:** the venues the agent can use (Polymarket and Kalshi).
- **Stop-loss:** the agent stops if the account falls this far below its highest value (20%).
- **Kill switch:** the operator turns this on to stop all trading immediately (off).

The policy also blocks an order when the agent does not have enough cash, or when a number in the order is wrong or missing. A safety stop blocks new buys. But the system can still sell or close positions, because that makes the risk smaller.

The strategy pack only proposes orders. The policy makes the final decision. Thus, a bad strategy cannot break the rules.

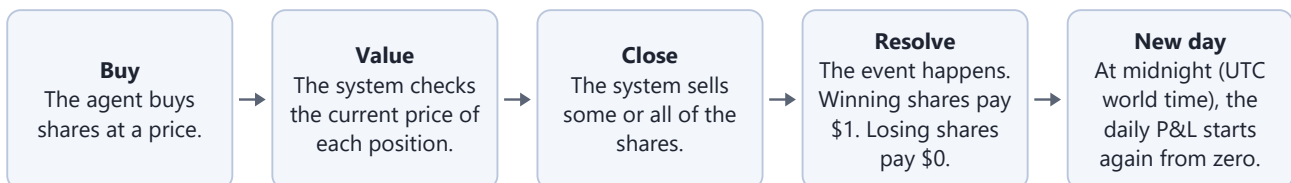
The main parts

The project has these parts. Each part is a separate program or a separate group of code.

Part	What it does	Folder
Agent loop	Runs each cycle on Chainlink CRE.	cre/agent-loop
Core logic	Holds the policy checks and the strategy pack. It also converts data from each venue into one common format.	packages/core
Control API	Keeps the policy, the practice account, the positions, the orders, and the audit log. It values the positions, closes them, and pays them out when a market ends.	services/control-api.ts
Database	Saves the data in SQLite files. The control API and the payment service use it. A restart does not lose money or orders.	packages/db
Payment service	Holds the agent's wallet. It pays for data, keeps a receipt for each payment, and watches each payment until the money reaches its final place.	services/cardano-agent.ts
Score provider	A test data seller. It charges 1 ADA for each set of scores.	services/score-provider.ts
Market feed	Reads Polymarket and Kalshi every 30 seconds. It does not use data older than 90 seconds.	services/market-feed.ts
Operator UI	A web page with three areas: Policy, Portfolio, and Audit. It updates every 2 seconds and marks each order PAPER or LIVE. It cannot change the policy or send orders.	ui

The life of a trade

The system now follows each paper trade from start to end:



- The system does not use a price that is more than 90 seconds old to close a position.
- When the system sells shares, the profit or loss on those shares becomes fixed.
- If a market is cancelled, the account gets back the money that it paid.
- The system saves a summary of each day.
- Each sale and each payout has its own key. If a request comes twice, the system does it only once.

Example. The command `bun run paper:demo` runs one full practice trade. The practice account started the day with \$1,000. It bought shares, sold part of them, and then the market ended. The account finished the day with \$1,004.40.

How the agent pays for data

The agent pays the data seller before it gets the scores. These are the steps:

- 1 The agent asks for scores. The data seller replies “payment required” and gives a price of 1 ADA.
- 2 The payment service saves a receipt first. Then it pays through Masumi. Masumi holds the money in escrow.
- 3 The payment service checks that the blockchain shows the payment.
- 4 The agent asks again with proof of payment. The data seller checks the proof and sends the scores.
- 5 The payment service saves the exact scores. If the agent asks again, it gets the same scores. It does not pay twice.
- 6 A background task watches the escrow. It records when the seller collects the money, when the agent gets a refund, or when there is a dispute. It keeps the evidence for each change.

The system keeps two facts separate: “the scores arrived” and “the money moved”. It shows that money moved only when the blockchain proves it.

At this time, the payments are simulated. The code for the Cardano test network is ready. Automatic tests check that code with recorded example replies. Nobody has made a real test-network payment yet.

How the project is being built

The building blocks

Tool	What we use it for
TypeScript	The programming language for all the main parts.
Bun	The tool that runs the TypeScript programs and the tests. It also installs code packages.
SQLite	A small database that keeps its data in one file.
Chainlink CRE	Runs the agent loop. Its command-line tool simulates the network on one computer.
Cardano, Masumi, and Blockfrost	Cardano carries the payments. Masumi holds them in escrow. Blockfrost is a service that reads data from Cardano.
Polymarket and Kalshi APIs	Give the market prices.
GitHub	Keeps the code and the history of each change. It also runs the automatic checks.

How a change gets into the project



The automatic checks include 324 tests. Type checks make sure that the parts of the code fit together. Style checks make sure that the code follows one style. On 5 October, all the checks pass on main.

The team has merged 22 pull requests. 14 of them came after the summary of 3 October.

People and AI assistants

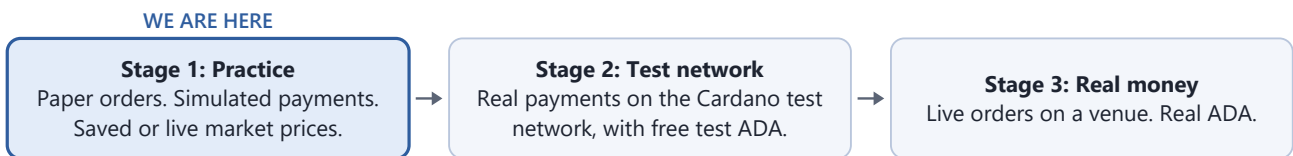
The team uses AI coding assistants to write much of the code. Examples are Cursor, Codex, and Claude Code. Team members give each assistant a task. Then they decide which changes to merge.

People and AI assistants follow one rule file, `AGENTS.md`. Some of its rules are:

- Send each change to main through a pull request.
- Write the docs in short, plain sentences.
- The operator UI only shows information. It cannot change the policy or send orders.
- Mark each order as paper or live, on the screen and in the docs.
- Never put secrets, such as passwords and keys, in the code.

Practice first, then real money

The project moves toward real money in three stages. Each stage adds more real money.



The project is at stage 1. The code for stage 2 is ready but not yet tried. Plan v2 makes stage 2 the next step and leaves stage 3 for later. By default, the payment service cannot connect to Cardano. Someone must turn on a setting first.

The plan for the hackathon is in `PLAN.md`. A proposed next step is in `PLAN_v2.md`.

What works now

- A full cycle runs from start to end in a CRE simulation. A simulation runs the program on one computer, in the same way that it will run on the real network.
- The agent reads real market data from Polymarket and Kalshi. It can also use saved data when no internet connection is available.
- The policy blocks politics markets. A market with an election or politics tag counts as politics, even when it also has other tags.
- The kill switch, the daily loss limit, and the stop-loss stop the agent before it buys data or sends orders.
- The practice account, the positions, the policy, and the orders stay safe after a restart. A repeated order does not fill twice.
- The control API can value, sell, and pay out paper positions. It keeps a daily P&L.
- The payment service keeps a receipt for each payment and tracks where the money goes.
- The operator UI updates every 2 seconds and fits a laptop window. If the control API stops, the UI shows a saved sample and labels it "Fixture data".

The folder docs/simulations/ keeps the records of three test runs from 3 October:

Test run	Result
Normal	The agent paid for scores, blocked 2 politics markets, and made 3 paper orders.
Kill switch on	The agent stopped before it bought data or sent orders.
Simulated loss of \$250	The daily loss limit and the stop-loss stopped the agent.

What is still simulated

- **All orders are paper orders.** No order goes to a venue.
- **All Cardano payments are simulated.** No payment goes to the blockchain.
- The payment message is our own local version of x402. It is not yet the standard x402 method on Cardano.
- The data seller is a test service. Its scores come from a fixed formula, not from real research.
- The wallet and its balance are demo values. The practice account starts with \$1,000.

What changed since 3 October

Pull request	Change
#10	The operator UI fits a laptop window.
#11	Politics tags take priority. Each audit record keeps its paper or live label.
#12	The agent does not buy data when no market can pass the policy. A blocked order does not take the place of an allowed order. The agent rejects a payment price that is wrong or too high.
#13	The automatic checks also test the operator UI and type-check the agent loop.
#14	The policy rejects wrong numbers and badly formed orders.
#15	Payment code for the Cardano test network and Masumi. It works without internet by default.
#17	Each payment is tied to the exact markets that the agent asked about.
#18	The database is now connected. The account and the orders stay safe after a restart. Paper positions can be valued, sold, and paid out.
#21, #22	The team made the code and the tests easier to read. The behavior did not change.
#23	The payment service tracks where the money goes after each payment.
#24	A proposed plan for the next stage (Plan v2).

Pull requests #19 and #20 changed the plan and then put it back. Pull request #16 was closed, and #18 replaced it.

What is not done yet

- Live orders on a venue. The hackathon plan permits paper orders, but each order must show which type it is.
- A real payment on the Cardano test network.
- A text box where the operator describes a strategy in plain words. One AI request would then change those words into strategy settings.
- Use of the real Chainlink CRE network. At this time, the agent runs only in simulation.

The team decided to keep these out of the hackathon: advanced order routing across venues, many strategy packs, and the Solana blockchain.

The next plan (Plan v2)

The team added a proposal for the next stage. Nothing in this section exists yet. The idea is a capital market for agents: give more money to the strategies that do well, inside strict limits.

- **Two strategy agents** with separate budgets. Strategy A buys a forecast and compares it with the market price. Strategy B compares the prices of the same event on different markets.
- **Paid research.** Strategy A pays for each forecast with a real payment on the Cardano test network.
- **A shared-event risk check.** Different markets can be about the same event. The check adds up the risk across those markets, so the agents cannot put too much money on one event.
- **A budget allocator.** After each round, it moves a small part of the budget to the strategy that did better. It keeps a cash reserve. It changes budgets only after it has enough results.
- **Outcome replay.** The system plays back recorded prices and results to score each strategy. Replay results do not prove real investment results.

The first step is one real payment on the Cardano test network.